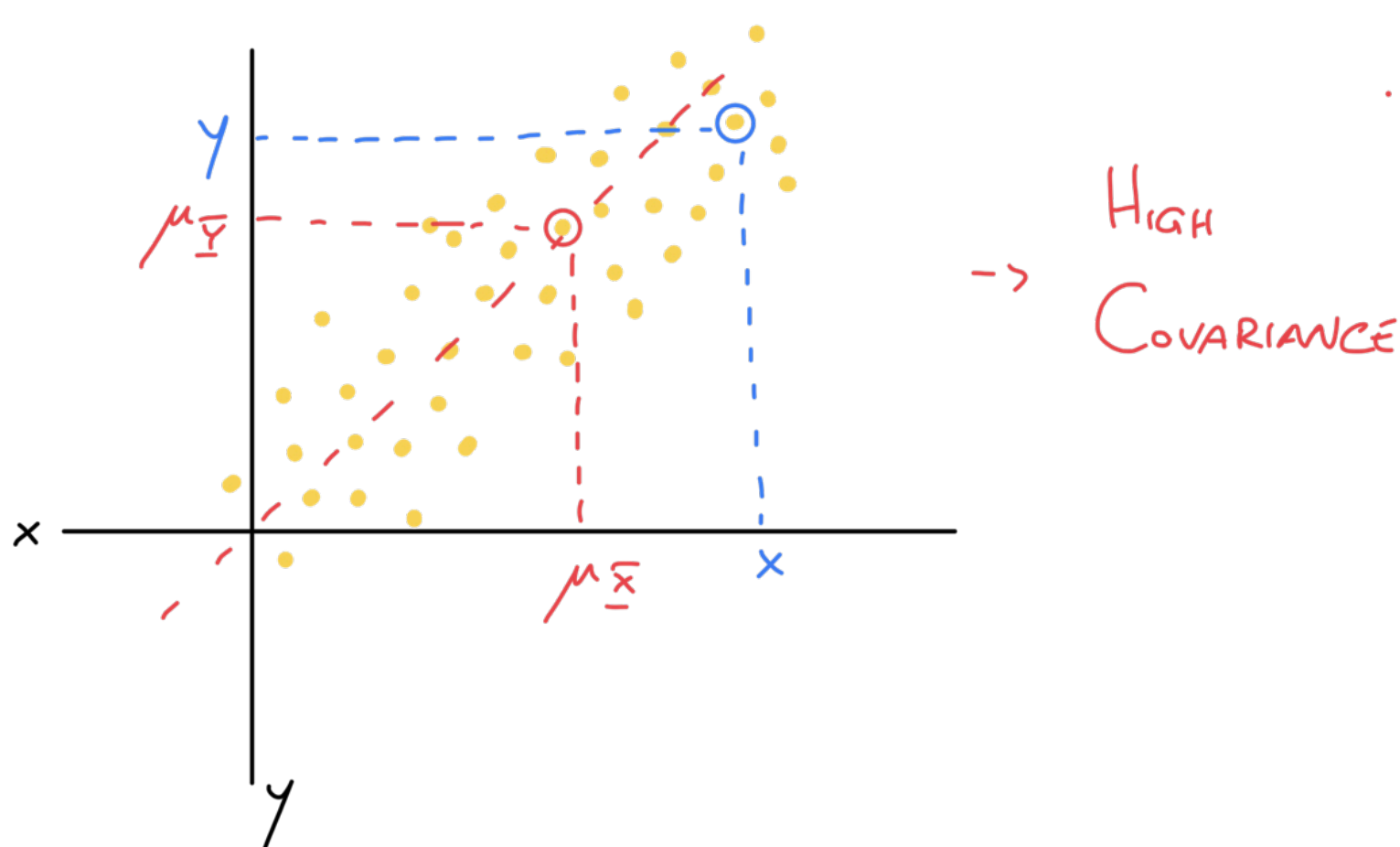


Covariance & Correlation

$$\text{Cov}(\bar{X}, \bar{Y}) = E\left(\underbrace{(\bar{X} - \mu_{\bar{X}})}_{E(\bar{X})} \underbrace{(\bar{Y} - \mu_{\bar{Y}})}_{E(\bar{Y})}\right)$$



PROPERTIES:

$$\begin{aligned}\text{Cov}(\bar{X}, \bar{Y}) &= E[\bar{X} \bar{Y} - \mu_x \bar{Y} - \mu_y \bar{X} + \mu_x \mu_y] \\ &= E(\bar{X} \bar{Y}) - \underbrace{\mu_x E(\bar{Y})}_{E(\bar{X})E(\bar{Y})} - \underbrace{\mu_y E(\bar{X})}_{E(\bar{Y})E(\bar{X})} + \underbrace{\mu_x \mu_y}_{E(\bar{X})E(\bar{Y})}\end{aligned}$$

$$\Rightarrow \boxed{\text{Cov}(\bar{X}, \bar{Y}) = E(\bar{X} \bar{Y}) - E(\bar{X})E(\bar{Y})}$$

$\Rightarrow$ If $\bar{X}, \bar{Y}$ ARE INDEPENDENT, THEN $\text{Cov}(\bar{X}, \bar{Y}) = 0$

$\hookrightarrow \text{Cov}(\bar{X}, \bar{Y}) = 0$ does NOT mean $\bar{X}, \bar{Y}$ indep!

$\Rightarrow \bar{X}$ uniform on $\{-1, 0, 1\}$

$$\bar{Y} = \bar{X}^2 \rightarrow \text{Cov}(\bar{X}, \bar{Y}) = 0$$

even though $\bar{X}, \bar{Y}$ are NOT independent!

$$\Rightarrow \text{Var}(\bar{X}) = \text{Cov}(\bar{X}, \bar{X})$$

$$\text{Var}(\bar{X} + \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) + 2\text{Cov}(\bar{X}, \bar{Y})$$

CORRELATION OF $\bar{X}, \bar{Y}$

$$\text{Corr}(\bar{X}, \bar{Y}) = \frac{\text{Cov}(\bar{X}, \bar{Y})}{\text{SD}(\bar{X}) \text{SD}(\bar{Y})} = \frac{\sigma_{\bar{X}\bar{Y}}}{\sigma_{\bar{X}} \sigma_{\bar{Y}}}$$

$\Rightarrow$ NORMALIZED COVARIANCE